

Lecture 10. Potential Outcomes & The Causal Inference Problem

ATE · ATT · ATU · Selection Bias · Randomization

BANA 4373 / ECON 4370 — Applied Data Tools for Economics & Business

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Introduction to Causal Inference

Today's Agenda

- 1 The Fundamental Problem of Causal Inference
- 2 Potential Outcomes & the Switching Equation
- 3 ATE, ATT, ATU — Three Parameters of Interest
- 4 Simple Difference in Outcomes (SDO) Decomposition
- 5 Selection Bias & Heterogeneous Treatment Effect Bias
- 6 Randomization as the Solution

Reference: Cunningham (2021), Causal Inference: The Mixtape, Ch. 4

How Lecture 10 Builds on Lecture 8

Last time: Regression

- OLS estimates correlations between X and Y
- Coefficients describe associations, not causes
- We saw omitted variables can flip signs
- **Question: when CAN regression identify a causal effect?**

Today: Causal Inference

- A formal framework for what "causal" means
- Why comparing treated vs. untreated is usually wrong
- A precise vocabulary: ATE, ATT, selection bias
- **Randomization as the gold standard**

The Fundamental Problem of Causal Inference

A causal effect compares two states of the world for the same unit:

Actual world

Patient takes aspirin $\rightarrow$ headache severity = 2

Counterfactual world

Same patient, no aspirin $\rightarrow$ headache severity = ?

The Problem: We only ever observe ONE state of the world per unit.

The counterfactual outcome is always missing. This is what Holland (1986) called *"the fundamental problem of causal inference."*

Potential Outcomes Notation

Each unit i has two potential outcomes (Neyman 1923; Rubin 1974):

Y^1_i

Outcome if unit i receives the treatment ($D = 1$)

Y^0_i

Outcome if unit i does NOT receive the treatment ($D = 0$)

$\delta_i = Y^1_i - Y^0_i$

Individual treatment effect for unit i

The Switching Equation

$$Y_i = D_i \cdot Y^1_i + (1 - D_i) \cdot Y^0_i$$

If $D=1 \rightarrow$ we observe Y^1 but Y^0 is missing. If $D=0 \rightarrow$ we observe Y^0 but Y^1 is missing.

Three Causal Parameters of Interest

ATE

Average Treatment Effect

Mean effect across the entire population

$$E[Y^1_i] - E[Y^0_i]$$

ATT

Average Treatment Effect on the Treated

Mean effect for those who actually received treatment

$$E[Y^1_i \mid D=1] - E[Y^0_i \mid D=1]$$

ATU

Average Treatment Effect on the Untreated

Mean effect for those who did NOT receive treatment

$$E[Y^1_i \mid D=0] - E[Y^0_i \mid D=0]$$

ATE

Average Treatment Effect

Mean effect across the entire population

$$ATE = \pi \cdot ATT + (1-\pi) \cdot ATU$$

π =share treated

Example: Surgery vs. Chemo — Full Potential Outcomes

Imagine we could observe BOTH potential outcomes (post-treatment life span in years):

Patient	Y^1 (Surgery)	Y^0 (Chemo)	$\delta_i = Y^1 - Y^0$
1	7	1	6
2	5	6	-1
3	5	1	4
4	7	8	-1
5	4	2	2
6	10	1	9
7	1	10	-9
8	5	6	-1
9	3	7	-4
10	9	8	1

Computing the ATE

Let's manually compute the ATE (we'll do it in Python later)

$$ATE = E[Y^1] - E[Y^0]$$

$E[Y^1]$ = the average outcome if everyone in the population receives treatment.

$E[Y^1] = (7+5+5+7+4+10+1+5+3+9)/10 = 5.6$ (if everyone gets surgery the average life span is 5.6 years)

$E[Y^0]$ = the average outcome if no one receives treatment

$E[Y^0] = (1+6+1+8+2+1+10+6+7+8)/10 = 5$ (if no one gets surgery and they all get chemo instead the average life span is 5 years).

ATE answers the question: on average, how much would outcomes change if we gave the treatment to everyone instead of no one?

Patient	Y^1 (Surgery)	Y^0 (Chemo)
1	7	1
2	5	6
3	5	1
4	7	8
5	4	2
6	10	1
7	1	10
8	5	6
9	3	7
10	9	8

$$ATE = E[Y^1] - E[Y^0] = 5.6 - 5.0 = 0.6 \rightarrow \text{on average, surgery increases life expectancy by 0.6 years compared to chemo.}$$

ATE's intuition

ATE is population level comparison (not individual).

The ATE compares two hypothetical worlds:

World 1: $E[Y^1]$: where everyone is treated

World 2: $E[Y^0]$: where no one is treated

Remember that for each individual, we never observe both outcomes, but the ATE average across the whole population

Key insight: ATE is the answer to a policy question: What would happen if we changed the treatment status for everyone?

The "Perfect Doctor" — What We Actually Observe

Now assume the perfect doctor who assigns each patient to their **best** treatment (surgery if $Y^1 > Y^0$, chemo if $Y^1 < Y^0$). After the assignment, we only see the observed outcome, the counterfactual vanishes:

Patient	Y (observed)	D
1	7	1 (surgery)
2	6	0 (chemo)
3	5	1 (surgery)
4	8	0 (chemo)
5	4	1 (surgery)
6	10	1 (surgery)
7	10	0 (chemo)
8	6	0 (chemo)
9	7	0 (chemo)
10	9	1 (surgery)

This is what happens in practice.

Patients are not randomly assigned, rather patients receive the treatment doctors believe is best for them.

We only observe one outcome per person.
The counterfactual is never observed.

So instead of seeing both outcomes, we only observe data like the table on the left.

Q: How do we now estimate the effect of the treatment?

A: We compare average outcomes across groups.

What is the Simple Difference in Outcomes(SDO)?

The comparison in average outcomes across groups is known as the Simple Difference in Outcomes (SDO). This is what we can compute from the data.

The SDO is defined as follows:

$SDO = E[Y \mid D = 1] - E[Y \mid D = 0]$  SDO = average outcome with treatment - average outcome w/out treatment

Rewrite SDO using the potential outcomes notation:

$$SDO = E[Y^1 \mid D = 1] - E[Y^0 \mid D = 0]$$

Add and subtract the missing counterfactual $E[Y^0 \mid D = 1]$ to get the following

$$SDO = \underbrace{(E[Y^1 \mid D = 1] - E[Y^0 \mid D = 1])}_{\text{Average Treatment on Treated (ATT)}} + \underbrace{(E[Y^0 \mid D = 1] - E[Y^0 \mid D = 0])}_{\text{Selection Bias}}$$

SDO and ATE

From the previous slide we obtain that: **SDO = AVERAGE TREATMENT ON THE TREATED (ATT) + SELECTION BIAS**

The simple difference in outcomes (SDO) gives us the treatment effect for the treated, plus any pre-existing differences between groups.

This definition of the SDO is useful but usually what we want is the ATE (the average effect for everyone). To obtain this we add and subtract ATE from the SDO definition above:

$$\text{SDO} = \text{ATE} + \text{Selection Bias} + \underbrace{\text{ATT} - \text{ATE}}_{\text{HTE Bias}} \rightarrow \text{SDO} = \text{ATE} + \text{Selection Bias} + \text{HTE Bias}$$

- **Selection Bias:** treated vs untreated differ even without treatment
- **HTE Bias:** people with larger gains are more likely to be treated

Even if the Selection Bias is zero, $\text{SDO} \neq \text{ATE}$ when treatment effects are heterogeneous

If people who benefit the more are more likely to be treated, then $\text{ATT} > \text{ATE}$ and the HTE Bias is positive

HTE BIAS, ATT and ATU

We can also express the HTE Bias as a function of the ATT and ATU.

From before we have that:

$$\text{HTE Bias} = \text{ATT} - \text{ATE}$$

Remember that:

$$\text{ATE} = \pi \cdot \text{ATT} + (1 - \pi) \cdot \text{ATU}$$

Substitute ATE:

$$\text{HTE} = \text{ATT} - [\pi \text{ ATT} + (1 - \pi) \text{ ATU}]$$

Simplify:

$$\text{HTE} = (1 - \pi)(\text{ATT} - \text{ATU})$$

Computing ATT

Let's use the potential outcomes table to compute ATT.

Patient	Y ¹ (Surgery)	Y ⁰ (Chemo)	D
1	7	1	1
2	5	6	0
3	5	1	1
4	7	8	0
5	4	2	1
6	10	1	1
7	1	10	0
8	5	6	0
9	3	7	0
10	9	8	1

$$ATT = E[Y^1 \mid D=1] - E[Y^0 \mid D=1]$$

ATT asks: for the patients who actually got surgery, what is the difference between their surgery outcome and their chemo outcome?

The surgery group is patients 1, 3, 5, 6, 10.

Step 1: Pull their Y¹ values (what actually happened): 7, 5, 4, 10, 9

$$\text{Average} = (7 + 5 + 4 + 10 + 9) / 5 = 35 / 5 = 7.0$$

Step 2: Pull their Y⁰ values (the counterfactual, what would have happened under chemo): 1, 1, 2, 1, 8

$$\text{Average} = (1 + 1 + 2 + 1 + 8) / 5 = 13 / 5 = 2.6$$

$$\text{Step 3: Subtract: } ATT = 7.0 - 2.6 = 4.4$$

Interpretation: For the patients the doctor chose to give surgery, surgery added 4.4 years of life on average compared to what chemo would have given them. The doctor picked well, these patients genuinely benefit from surgery.

Computing ATU

Let's use the potential outcomes table to compute ATU.

Patient	Y ¹ (Surgery)	Y ⁰ (Chemo)	D
1	7	1	1
2	5	6	0
3	5	1	1
4	7	8	0
5	4	2	1
6	10	1	1
7	1	10	0
8	5	6	0
9	3	7	0
10	9	8	1

$$ATU = E[Y^1 | D=0] - E[Y^0 | D=0]$$

ATU asks: for the patients who actually got chemo, what would have happened if they had gotten surgery instead? The surgery group is patients 1, 3, 5, 6, 10.

Step 1: Pull their Y¹ values (the counterfactual, what would have happened under surgery): 5, 7, 1, 5, 3

Average = $(5 + 7 + 1 + 5 + 3) / 5 = 21 / 5 = 4.2$

Step 2: Pull their Y⁰ values (what actually happened): 6, 8, 10, 6, 7

Average = $(6 + 8 + 10 + 6 + 7) / 5 = 37 / 5 = 7.4$

Step 3: Subtract: **ATU = 4.2 - 7.4 = -3.1**

Interpretation: For the patients the doctor chose to give chemo, surgery would have reduced their lifespan by 3.2 years on average. Again, the doctor picked well, these patients are genuinely better off on chemo.

Verify ATT and ATU

In slide 8 we computed the ATE directly to be -0.6

Now, let's verify and show that the ATE can be computed as a Weighted Average between ATT and ATU

$$\begin{aligned} \text{ATE} &= \pi \cdot \text{ATT} + (1 - \pi) \cdot \text{ATU} \\ &= 0.5 \times 4.4 + 0.5 \times (-3.2) \\ &= 2.2 + (-1.6) \end{aligned}$$

$$\text{ATE} = 0.6 \quad \checkmark$$

Notice: The ATT and ATU have opposite signs. Surgery helps some people and hurts others. The ATE of 0.6 is a mild positive average, but it masks enormous heterogeneity underneath.

The "Perfect Doctor" and the SDO

Let's continue with the perfect doctor who assigns each patient to their **best** treatment (surgery if $Y^1 > Y^0$, chemo if $Y^1 < Y^0$) and compute the SDO.

Patient	Y (observed)	D
1	7	1 (surgery)
2	6	0 (chemo)
3	5	1 (surgery)
4	8	0 (chemo)
5	4	1 (surgery)
6	10	1 (surgery)
7	10	0 (chemo)
8	6	0 (chemo)
9	7	0 (chemo)
10	9	1 (surgery)

Surgery group mean : $(7+5+4+10+9)/5 = 7.0$

Chemo group mean: $(6+8+10+6+7)/5 = 7.4$

SDO = 7.0 - 7.4 = -0.4

True ATE: **0.6**

ATT: **4.4**

ATU: **-3.2**

The SDO Decomposition

The simple difference in outcomes contains THREE components:

SDO = $E[Y^1 | D=1] - E[Y^0 | D=0]$ *What we can compute from data*

= ATE $E[Y^1] - E[Y^0]$ *← What we WANT*

+ Selection Bias $E[Y^0 | D=1] - E[Y^0 | D=0]$ *← Baseline difference between groups*

+ HTE Bias $(1-\pi)(ATT - ATU)$ *← Different returns to treatment*

What Is Selection Bias?

$$\text{Selection Bias} = E[Y^0 \mid D=1] - E[Y^0 \mid D=0]$$

The difference between the treatment and control groups that would exist even if neither group received treatment. It measures how the groups differ at baseline.

In our example:

$E[Y^0 \mid D=1] = 2.6$ (chemo outcome for the surgery group: $(1 + 1 + 2 + 1 + 8) / 5 = 13 / 5 = 2.6$)

$E[Y^0 \mid D=0] = 7.4$ (chemo outcome for the chemo group: $(6 + 8 + 10 + 6 + 7) / 5 = 37 / 5 = 7.4$)

Selection Bias = $2.6 - 7.4 = -4.8$

Intuition: The doctor sent patients with terrible chemo prognosis (averaging only 2.6 years) to chemo. The chemo group has a much better baseline under chemo (7.4 years). The -4.8 gap drags the SDO down and has nothing to do with how well surgery works, it is a pre-existing difference between the groups.

What Is Heterogeneous Treatment Effect Bias?

$$\text{HTE Bias} = (1 - \pi)(\text{ATT} - \text{ATU})$$

This bias exists when the treatment effect differs between people who chose to be treated and those who did not, and the target parameter is ATE, not ATT.

In our example:

ATT = 4.4 (surgery helps the surgery group a lot)

ATU = -3.2 (surgery would HURT the chemo group)

$\pi = 0.5$ (half in each group)

$$\text{HTE Bias} = (1 - 0.5)(4.4 - (-3.2)) = 0.5 \times 7.6 = 3.8$$

Key insight: Even without selection bias, if we target ATE but $\text{ATT} \neq \text{ATU}$, we still get the wrong answer. People sort into treatment based on their expected gains. The 7.6 point gap in treatment effects, weighted by the untreated share inflates the SDO upward by 3.8.

Summary

Final Check: Putting It All Together

$SDO = ATE + \text{Selection Bias} + \text{HTE Bias}$

$-0.4 = 0.6 + (-4.8) + 3.8$

$-0.4 = -0.4 \checkmark$



The true effect of surgery is positive (0.6), but the massive negative selection bias (-4.8) overwhelms it. The HTE bias (+3.8) partially offsets the selection bias but not enough. The net result is an SDO of -0.4 that wrongly suggests surgery is harmful.

Component	Formula	Value
ATE	$E[Y^1] - E[Y^0]$	0.6
ATT	$E[Y^1 D=1] - E[Y^0 D=1]$	4.4
ATU	$E[Y^1 D=0] - E[Y^0 D=0]$	-3.2
Selection Bias	$E[Y^0 D=1] - E[Y^0 D=0]$	-4.8
HTE Bias	$(1-\pi)(ATT - ATU)$	3.8
SDO	$ATE + \text{Selection Bias} + \text{HTE Bias}$ $ATT + \text{Selection Bias}$	-0.4

Why Naïve Comparisons Fail: The Economics

Economic reasoning tells us that selection bias is nearly inevitable in observational data:

Job training

Workers who enroll may be more motivated — they'd do better even without the program.

College premium

Students who attend college differ in ability, family resources, ambition from those who don't.

Hospital quality

Sicker patients go to better hospitals — comparing outcomes makes good hospitals look bad.

Minimum wage

States that raise wages differ economically from states that don't.

Rational choice always pushes against the independence assumption.

The Independence Assumption

$$(Y^1, Y^0) \perp D$$

Treatment assignment is independent of potential outcomes — treatment was assigned for reasons having NOTHING to do with expected gains.

Under independence:

Selection bias = 0

$$E[Y^0 | D=1] - E[Y^0 | D=0] = 0$$

HTE bias = 0

$$ATT = ATU \rightarrow (1-\pi)(ATT-ATU) = 0$$

Therefore: $SDO = ATE$ ← Simple comparison IS the causal effect!

Randomization: The Gold Standard

How do we guarantee independence? Flip a coin.

Physical randomization (coin flip, random number generator) ensures that treatment assignment is determined by chance, not by patient characteristics, doctor judgment, or self-selection.

Because the assignment mechanism is known and controlled by the researcher, we can guarantee that (on average), in the population treated and control groups are identical on ALL dimensions, observed and unobserved.

Landmark randomized experiments:

1954	Salk Polio Vaccine Trial	200K+ children randomized → proved vaccine works
1971–82	RAND Health Insurance	7,700 people → cost-sharing reduces utilization
2008	Thornton HIV Study	Cash incentives → 2× more likely to learn HIV status

Randomization vs No Randomization

When randomization is possible:

$$\text{SDO} = \text{ATE}$$

No Selection Bias and No HTE Bias

When randomization is NOT possible

We must construct a credible counterfactual:

1. Differences-in-Differences (DiD): build counterfactual using trends
2. Instrumental Variables (IV): exogenous variation in treatment
3. Synthetic Control: construct a comparison unit

All causal inference methods try to replicate randomization

SUTVA: The Fine Print on Potential Outcomes

The Stable Unit Treatment Value Assumption (Rubin) requires three things:

1. No variation in treatment doses

Every treated unit gets the same "dose" — e.g., all surgeons are equally skilled.

2. No spillovers / externalities

Treating unit i doesn't change unit j 's outcome — e.g., vaccinating me doesn't protect you (herd immunity violates this!).

3. No general equilibrium effects

Scaling up the treatment doesn't change the environment — e.g., sending everyone to college would change the wage premium.

Key Takeaways

- 1 A causal effect is the difference between two potential outcomes for the SAME unit — but we only ever observe one.
- 2 The simple difference in outcomes (SDO) = ATE + Selection Bias + HTE Bias. It almost never equals the causal effect in observational data.
- 3 Selection bias arises because treated and control groups differ at baseline — people choose treatments based on expected gains.
- 4 Randomization eliminates both selection bias and HTE bias by making treatment assignment independent of potential outcomes.
- 5 SUTVA reminds us: no dose variation, no spillovers, no general equilibrium effects.

Questions?

Next: Open the Lecture 10 notebook

We'll compute ATE, ATT, ATU, selection bias, and HTE bias hands-on in Python.
Then we'll simulate randomization to see both biases vanish.